Exercise 3: Hypothesis Test for a New Teaching Method

Setup: let $\mu_{diff} = \mu_{new} - \mu_{traditional}$ be the true mean difference in exam scores between the new and the traditional teaching method. In the sample, the new method has mean 78 and the traditional method has mean 75 (observed difference 3), and the test gave $p = 0.12$ at significance level $\alpha = 0.05$.

1. Null and alternative hypotheses

$$\mathbf{H_0}: \mu_{diff} = 0 \quad \text{(the two methods have the same true mean score)}$$

$$\mathbf{H_1}: \mu_{diff} \neq 0 \quad \text{(the true mean scores differ)}$$

The question asks whether the new method leads to a *different* average score, so H_1 does not specify a direction.

2. Type of test

A two-sided test should be used. Reasons:

• The research question is whether the average score is *different*, not whether it is higher or lower. A two-sided alternative ($\mu_{diff} \neq 0$) matches this wording exactly.

• A new method could plausibly help *or* hurt. A two-sided test detects a deviation from H_0 in either direction, whereas a one-sided test would ignore effects in the other direction.

• The direction of the test must be fixed from the research question *before* looking at the data. Choosing a one-sided test just because the sample difference happens to be positive ($78 > 75$) would bias the test in favor of rejecting H_0.

Conclusion of the test

Since $p = 0.12 > \alpha = 0.05$, we **fail to reject $\mathbf{H_0}$**. There is not enough evidence at the 5% level that the new method changes the mean exam score. The observed 3-point difference is consistent with random sampling variation. (This does not prove the methods are equal; it only means the data do not provide sufficient evidence of a difference.)